

Simulation and Spectroscopic Analysis of a Superconducting Transmon Qubit Coupled to a Microwave Resonator

A Dissertation Submitted in Partial Fulfilment of the Requirements
for the Degree of

Master of Technology

by

Prachi Tyagi

SR No. 24255



under the guidance of

Prof. Baladitya Suri

DEPARTMENT OF INSTRUMENTATION AND APPLIED PHYSICS
INDIAN INSTITUTE OF SCIENCE
BENGALURU - 560012

May 2026

Abstract

This report presents the simulation and spectroscopic analysis of a superconducting transmon qubit coupled to a microwave resonator in the circuit QED architecture. The study focuses on both frequency-domain and time-domain characterization of the qubit–resonator system using numerical simulations based on the Lindblad master equation.

In the frequency domain, resonator and qubit spectroscopy were performed using both brute-force and adaptive methods. A phase-asymmetric Lorentzian model was used to extract key resonator parameters such as resonance frequency, linewidth, and asymmetry. To improve computational efficiency, an adaptive spectroscopy algorithm was implemented, which significantly reduces redundant frequency sampling while preserving spectral accuracy.

Photon-number splitting was observed in the two-tone spectroscopy simulations, confirming the strong dispersive interaction between the qubit and the resonator. Power-dependent spectroscopy further revealed linewidth broadening at higher drive powers, consistent with power broadening effects arising from increased qubit excitation and saturation. The evolution of spectral features with increasing drive power was analyzed, showing clear transitions between low-power linear, intermediate nonlinear, and high-power saturation regimes.

In the time domain, Rabi oscillations, Ramsey interference, and energy relaxation (T_1) measurements were simulated to extract qubit coherence properties. The dephasing time (T_2^*) and relaxation time (T_1) were obtained using standard fitting models, demonstrating expected exponential decay behavior.

Additionally, optimization techniques such as knee-point detection for readout power calibration and binary search for optimal qubit drive power were implemented to ensure stable and efficient operating conditions.

Overall, this work provides a comprehensive numerical framework for analyzing transmon qubit dynamics, highlighting the interplay between measurement precision, computational efficiency, and nonlinear effects in circuit QED systems.

Contents

1	Introduction	6
1.1	Quantum Computing and Qubits	6
1.2	Superconducting Qubits and Circuit QED	7
1.3	Qubit Characterization and Spectroscopy	7
1.4	Overview of the Present Work	8
2	Theory	9
2.1	Circuit Quantum Electrodynamics and Superconducting Qubits . .	9
2.2	Resonator Dynamics and Open Quantum Systems	10
2.3	Jaynes–Cummings Model and Dispersive Regime	12
2.4	Input-Output Theory and S_{21} Measurements	13
2.5	Two-Tone Spectroscopy	13
2.6	Photon-Number Splitting	14
2.6.1	Power Broadening	15
2.7	Anharmonicity of the Transmon Qubit	16
2.8	Time-Domain Characterization of Superconducting Qubits	18
2.8.1	Bloch Sphere Representation	18
2.8.2	Rabi Oscillations	18
2.8.3	Ramsey Interference and Dephasing Time	19
2.8.4	Relaxation Time and Energy Decay	19
2.8.5	Relaxation and Decoherence	19
3	Experimental Setup	21
3.1	Overview of Experimental Setup	21
3.2	Microwave Input Lines and Attenuators	21
3.3	Vector Network Analyzer (VNA)	21
3.4	Signal Generator (SG)	22
3.5	Cryogenic Measurement System	22
4	Measurement Methodology	23
4.1	Experimental Methodology	23

4.1.1	Resonator Spectroscopy Measurements	23
4.1.2	Coarse Frequency Sweep	24
4.1.3	Naive Sweep	24
4.1.4	Adaptive Spectroscopy Algorithm	24
4.1.5	Signal Processing and Filtering	24
4.1.6	Phase-Asymmetric Lorentzian Fitting	24
4.1.7	Readout Power Calibration via Transition Detection	25
4.2	Methodology and Simulation	25
4.2.1	Overview of Simulation Framework	25
4.2.2	System Model	26
4.2.3	Simulation Parameters	26
4.2.4	Bare Resonator Spectroscopy	27
4.2.5	Qubit-Resonator System Simulation	27
4.2.6	Two-Tone Spectroscopy Simulation	27
4.2.7	Photon-Number Splitting Simulation	27
4.2.8	Power Broadening Simulation	28
4.2.9	Optimal Power Estimation	28
4.2.10	Transmon Anharmonicity Simulation	29
4.3	Time-Domain Measurement Simulations	31
4.3.1	Rabi Oscillation Simulation	31
4.3.2	Ramsey Interference Simulation	31
4.3.3	T_1 Relaxation Measurement	31
5	Results and Discussion	32
5.1	Resonator Spectroscopy Results	32
5.1.1	Coarse and Adaptive Resonator Spectroscopy	32
5.1.2	Power Dependence of Resonator Response	33
5.1.3	Observation of Photon-Number Splitting	33
5.1.4	Power Broadening Analysis	34
5.1.5	Power Dependence of Peak Height and Linewidth	36
5.1.6	Anharmonicity Results	37
6	Conclusion and Future Work	40
6.1	Future Work	40
A	Additional Methods and Algorithms	41
A.1	Knee Point Detection for Optimal Readout Power	41
A.2	Binary Search for Optimal Drive Power	41
A.3	Brute Force Spectroscopy Method	42
A.4	Adaptive Spectroscopy Algorithm	42

A.5	Lorentzian Fitting Model	42
A.6	Simulation Framework Details	42
A.7	Time Domain Results	43

List of Figures

2.1	Schematic representation of a Josephson junction.	10
2.2	Comparison between a harmonic oscillator and a transmon qubit energy spectrum showing unequal level spacing and anharmonicity.	17
3.1	Overall experimental setup and cryogenic measurement system used for superconducting qubit characterization.	22
5.1	Evolution of photon-number splitting in two-tone spectroscopy with increasing readout power. Low-power spectra show a small number of resolved photon-number peaks, while intermediate powers reveal multiple equally spaced transitions separated by approximately 2χ . At high power, nonlinear broadening and overlap of spectral features indicate the onset of breakdown of the simple dispersive approximation.	34
5.2	Power broadening effects and linewidth dependence on drive power showing spectral broadening and transition saturation behavior.	36
5.3	Brute-force spectroscopy results at low (-100 dBm), intermediate (-80 dBm), and high (-60 dBm) drive powers. The higher-level f_{12} transition gradually emerges with increasing drive power, while strong power broadening appears at high drive strength.	38
5.4	Adaptive spectroscopy results at low (-100 dBm), intermediate (-80 dBm), and high (-60 dBm) drive powers, showing efficient resolution of the f_{01} and f_{12} transitions with increasing drive strength.	38
A.1	Additional time-domain measurement results obtained from the transmon qubit simulations.	43

Chapter 1

Introduction

1.1 Quantum Computing and Qubits

Quantum computing is an emerging field that uses the principles of quantum mechanics to process information. Unlike classical computers, which use bits that exist only in the states 0 or 1, quantum computers use quantum bits or qubits. A qubit can exist in a superposition of states, allowing quantum systems to perform certain computational tasks more efficiently than classical systems.

The state of a qubit can be written as

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle \quad (1.1)$$

where α and β are complex probability amplitudes satisfying

$$|\alpha|^2 + |\beta|^2 = 1 \quad (1.2)$$

The quantities $|\alpha|^2$ and $|\beta|^2$ represent the probabilities of measuring the states $|0\rangle$ and $|1\rangle$, respectively. Quantum systems also exhibit entanglement, where the states of multiple qubits become correlated in a way that cannot be described independently. Such properties form the basis of quantum algorithms and quantum information processing.

A major challenge in quantum computing is preserving coherence of the qubit states. Interaction with the surrounding environment introduces noise and decoherence, which affects the reliability of quantum operations. Considerable research is therefore focused on developing stable qubit platforms with improved coherence properties.

Different physical systems have been explored for qubit implementation, including trapped ions, photonic systems, semiconductor spin systems, and superconducting circuits. Among these approaches, superconducting qubits have

emerged as one of the leading platforms for scalable quantum computing.

1.2 Superconducting Qubits and Circuit QED

Superconducting qubits are artificial quantum systems fabricated using superconducting electrical circuits operating at millikelvin temperatures. At these temperatures, electrical resistance becomes negligible and quantum effects can be observed at the circuit level.

The fundamental element used in superconducting quantum circuits is the Josephson junction, which consists of two superconductors separated by a thin insulating barrier. The Josephson junction introduces nonlinearity into the circuit and allows the formation of discrete energy levels required for qubit operation.

Among various superconducting qubit architectures, the transmon qubit is widely used because of its reduced sensitivity to charge noise and relatively long coherence times. The transmon is designed by increasing the ratio of Josephson energy to charging energy, thereby improving qubit stability while maintaining sufficient anharmonicity for selective quantum control.

In circuit quantum electrodynamics (circuit QED), superconducting qubits are coupled to microwave resonators. The resonator is used for qubit control and readout, while the qubit-resonator interaction enables different spectroscopy and characterization measurements. Circuit QED has become an important framework for studying light-matter interaction in superconducting quantum systems and forms the basis of many modern quantum computing architectures.

1.3 Qubit Characterization and Spectroscopy

Characterization of superconducting qubits is necessary for understanding the behavior of the coupled qubit-resonator system. Different measurement techniques are used to extract important parameters such as resonator frequency, qubit transition frequency, coherence times, and coupling strengths.

One of the commonly measured quantities in superconducting quantum circuits is the microwave transmission coefficient S_{21} . By studying the variation of S_{21} with frequency, the spectral response of the resonator and the coupled qubit system can be analyzed.

Resonator spectroscopy is generally performed first to determine the resonance frequency of the cavity. When the qubit and resonator operate in the dispersive regime, the resonator response becomes dependent on the qubit state. This allows indirect measurement of the qubit through the resonator transmission spectrum.

After identifying the resonator characteristics, qubit spectroscopy is carried out to determine the transition frequency of the qubit. At low microwave drive powers, the qubit transition can be measured with minimal disturbance to the system. Increasing the drive power leads to additional nonlinear effects in the spectrum.

One such effect is photon-number splitting, where multiple spectral peaks appear due to the interaction between the qubit and quantized cavity photon states. Another important effect is power broadening, in which the linewidth of the spectral peak increases with increasing microwave drive power. Under strong driving conditions, dressed-state features may also appear due to hybridization between qubit states and cavity photon states.

Spectroscopy measurements are also useful for determining the anharmonicity of the transmon qubit. Unlike a harmonic oscillator with equally spaced energy levels, the transmon possesses unequally spaced transitions. This anharmonicity allows selective control of the lowest two energy states and enables the transmon to function effectively as a qubit.

In addition to frequency-domain measurements, time-domain experiments are used to study qubit dynamics and coherence properties. Commonly used techniques include Rabi oscillation measurements, Ramsey interference experiments, and relaxation measurements for extracting the coherence parameters T_1 and T_2 .

1.4 Overview of the Present Work

This work focuses on simulation-based characterization of a superconducting transmon-resonator system using techniques commonly employed in circuit QED experiments. The primary objective is to study the spectral response of the coupled system through frequency-domain analysis.

The characterization procedure includes resonator spectroscopy, dispersive regime identification, low-power qubit spectroscopy, photon-number splitting analysis, power broadening analysis, dressed-state observation, and extraction of transmon anharmonicity. The microwave transmission response of the system is analyzed using the S_{21} parameter under different operating conditions.

In addition to frequency-domain studies, a brief time-domain analysis is also performed to investigate coherent qubit dynamics. Simulations such as Rabi oscillations, Ramsey measurements, and relaxation experiments are considered to study coherence behavior and energy relaxation processes.

Overall, this work presents a computational study of spectroscopy and characterization techniques relevant to superconducting quantum hardware systems.

Chapter 2

Theory

2.1 Circuit Quantum Electrodynamics and Superconducting Qubits

Circuit quantum electrodynamics (circuit QED) is the superconducting circuit analogue of cavity quantum electrodynamics, where a quantum two-level system interacts with a confined electromagnetic field. In circuit QED architectures, superconducting qubits are coupled to microwave resonators that act as on-chip cavities for storing and manipulating microwave photons. A simplified circuit QED setup used in the experimental measurement chain is shown in Figure 3.1. The system consists of a superconducting qubit coupled to a coplanar waveguide resonator. Microwave signals are applied through the input port of the resonator, while the transmitted signal is measured at the output using microwave transmission techniques.

The qubit is generally positioned near the maximum electric field region of the resonator to enhance the interaction strength between the microwave field and the qubit. This interaction modifies the spectral response of the coupled system and enables qubit control and readout through resonator transmission measurements.

The resonator field is quantized and can be described in terms of photons. The energy levels of the resonator are given by

$$E_n = \hbar\omega_r \left(n + \frac{1}{2} \right) \quad (2.1)$$

where ω_r is the resonator frequency, n is the photon number, and $\hbar$ is the reduced Planck constant.

Among different qubit implementations, superconducting circuits have emerged as one of the leading platforms for scalable quantum computation. These circuits operate at millikelvin temperatures where superconductivity eliminates electrical

resistance and allows coherent quantum behavior to be observed.

The key nonlinear element used in superconducting quantum circuits is the Josephson junction, which consists of two superconductors separated by a thin insulating barrier.

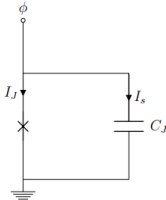


Figure 2.1: Schematic representation of a Josephson junction.

The Josephson junction provides the nonlinearity required for generating discrete energy levels in superconducting circuits. In the absence of this nonlinearity, the circuit behaves as a harmonic oscillator with equally spaced energy levels.

By combining Josephson junctions with capacitive and inductive circuit elements, artificial atoms with quantized energy levels can be engineered. The two lowest energy states are typically used as the computational basis states $|0\rangle$ and $|1\rangle$.

Among various superconducting qubit designs, the transmon qubit is one of the most widely used architectures. The transmon is derived from the Cooper-pair box and is designed to reduce sensitivity to charge noise by increasing the ratio of Josephson energy to charging energy. This improves coherence while preserving sufficient anharmonicity for selective qubit control.

Although the present simulation work focuses on transmon-based systems, other superconducting qubit architectures such as fluxonium are also actively studied because of their improved coherence properties in certain operating regimes.

2.2 Resonator Dynamics and Open Quantum Systems

In circuit QED experiments, the microwave resonator plays an important role in both qubit readout and spectroscopy measurements. The resonator is commonly modeled as a linear quantum harmonic oscillator with Hamiltonian

$$H = \hbar\omega_r a^\dagger a \quad (2.2)$$

where ω_r is the resonator frequency, and $a^\dagger$ and a are the photon creation and annihilation operators, respectively.

This Hamiltonian describes equally spaced energy levels characteristic of a harmonic oscillator. Resonator spectroscopy is typically performed by transmitting a microwave signal through the cavity and measuring the transmission response as a function of frequency. The resonance frequency is identified from the Lorentzian-shaped transmission spectrum.

In practical experiments, the resonator is continuously driven using external microwave fields. To simplify the description of the driven system, the Hamiltonian is often written in the rotating frame of the applied drive frequency as

$$H = \Delta a^\dagger a + \epsilon(a + a^\dagger) \quad (2.3)$$

where $\Delta = \omega_r - \omega_d$ represents the detuning between the resonator and drive frequencies, and ϵ is the drive amplitude.

Real superconducting systems are not perfectly isolated from the surrounding environment. As a result, dissipation and decoherence must be included while describing the system dynamics. These effects are commonly modeled using the Lindblad master equation

$$\frac{d\rho}{dt} = -i[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right) \quad (2.4)$$

where ρ is the density matrix of the system, H is the system Hamiltonian, and L_k are collapse operators describing different dissipation channels.

For resonator photon loss, the collapse operator is written as

$$L = \sqrt{\kappa} a \quad (2.5)$$

where κ denotes the resonator decay rate. The decay rate is related to the resonator quality factor through

$$\kappa = \frac{\omega_r}{Q} \quad (2.6)$$

where Q is the quality factor of the resonator.

Experimentally, the internal cavity field is not measured directly. Instead, microwave transmission measurements are performed using input-output theory. The transmitted signal is characterized using the scattering parameter S_{21} , which provides information about the resonator response, linewidth, and qubit-resonator interaction.

The resonator response additionally depends on the applied microwave drive power, which determines the average photon population inside the cavity. The intracavity photon number can be estimated using

$$n = \left(\frac{2\epsilon}{\kappa} \right)^2 \quad (2.7)$$

where n is the average photon number inside the resonator. Photon-number calibration is important because many nonlinear effects in superconducting systems depend directly on the number of photons inside the resonator.

2.3 Jaynes–Cummings Model and Dispersive Regime

To describe the interaction between the superconducting qubit and the resonator field, the coupled system is modeled using the Jaynes–Cummings Hamiltonian

$$H = \omega_c a^\dagger a + \frac{\omega_q}{2} \sigma_z + g(a^\dagger \sigma_- + a \sigma_+) \quad (2.8)$$

where ω_c is the cavity frequency, ω_q is the qubit transition frequency, and g is the coupling strength between the qubit and the resonator. The operators σ_z , σ_+ , and σ_- describe the qubit degrees of freedom.

The Jaynes–Cummings model captures the coherent exchange of energy between resonator photons and the superconducting qubit and forms the theoretical basis for circuit QED systems.

In most superconducting qubit experiments, the qubit and resonator operate in the dispersive regime, where their frequencies are sufficiently detuned from each other. Under this condition, direct energy exchange between the qubit and resonator is suppressed, while the resonator frequency acquires a qubit-state-dependent shift.

The effective resonator frequency in the dispersive regime can be expressed as

$$\omega_r^{\text{eff}} = \omega_r + \chi \langle \sigma_z \rangle \quad (2.9)$$

where χ denotes the dispersive shift and $\langle \sigma_z \rangle$ represents the qubit population.

This relation shows that the resonator transmission spectrum depends on the quantum state of the qubit. When the qubit occupies different energy states, the resonator response shifts accordingly, enabling indirect qubit measurement through microwave transmission.

At low drive powers, the qubit remains predominantly in the ground state and the shifted resonator response can be clearly observed. As the microwave power increases, the photon population inside the resonator also increases, leading to saturation effects and modifications in the transmission spectrum.

Under strong driving conditions, nonlinear phenomena such as photon-number splitting, dressed-state formation, and power broadening become observable. These

effects provide important insight into the dynamics of the coupled qubit–resonator system and form the basis of many spectroscopy measurements in superconducting quantum hardware. The dispersive regime plays an important role in superconducting qubit readout and frequency-domain characterization experiments.

2.4 Input-Output Theory and S_{21} Measurements

In superconducting circuit QED experiments, resonator characterization is commonly performed using microwave transmission measurements. In experimental setups, a Vector Network Analyzer (VNA) is typically used to measure the transmitted microwave response of the resonator.

Instead of directly probing the internal cavity field, the transmitted signal is analyzed using input-output theory. The transmission response is characterized through the transmission coefficient S_{21} , which describes the microwave transmission through the resonator as a function of frequency.

The measured S_{21} spectrum provides information about the resonator response and the coupled qubit–resonator interaction. Power-dependent transmission measurements additionally reveal nonlinear effects arising in the coupled qubit–resonator system.

2.5 Two-Tone Spectroscopy

In superconducting qubit systems, the qubit is not measured directly. Instead, it is probed indirectly through its interaction with the resonator. This is achieved using a technique known as two-tone spectroscopy.

In this method, two microwave signals are applied simultaneously:

- a fixed readout tone at the resonator frequency,
- a second tone that sweeps across a range of frequencies to excite the qubit.

The readout tone continuously probes the resonator, while the drive tone interacts with the qubit. When the drive frequency matches the qubit transition frequency, the qubit gets excited, which in turn modifies the resonator response due to dispersive coupling.

This change in the resonator transmission is detected as a dip or peak in the measured signal, allowing identification of the qubit frequency.

In the dispersive regime, the resonator response depends on the qubit state. As the applied microwave power increases, the coupled qubit–resonator system begins to exhibit several nonlinear effects in the spectroscopy spectrum. These

include photon-number splitting, dressed-state formation, and power-dependent broadening of the qubit transition linewidth.

2.6 Photon-Number Splitting

In circuit QED, a superconducting qubit coupled to a microwave resonator exhibits a phenomenon known as photon-number splitting. In the dispersive regime, where the detuning between the qubit and resonator is much larger than the coupling strength,

$$|\Delta| \gg g \quad (2.10)$$

the qubit and resonator no longer exchange energy directly. Instead, the resonator photons shift the qubit transition frequency.

The effective dispersive Hamiltonian is given by

$$H_{\text{disp}} = \omega_c a^\dagger a + \frac{\omega_q}{2} \sigma_z + \chi a^\dagger a \sigma_z \quad (2.11)$$

where ω_c is the resonator frequency, ω_q is the qubit frequency, $a^\dagger a$ represents the photon number operator, and χ is the dispersive shift. The interaction term $\chi a^\dagger a \sigma_z$ causes the qubit frequency to depend on the number of photons inside the resonator.

As a result, the qubit transition frequency becomes

$$\omega_q(n) = \omega_q + 2\chi n \quad (2.12)$$

where n denotes the intracavity photon number. This relation shows that each additional photon shifts the qubit frequency by an amount 2χ .

At low readout power, the resonator contains only a small number of photons, and the spectroscopy spectrum typically shows a single transition peak. However, as the readout power increases, the resonator enters a coherent state containing multiple photon number states simultaneously. Each photon occupation state produces a different qubit transition frequency, resulting in multiple equally spaced peaks in the spectroscopy spectrum corresponding to

$$n = 0, 1, 2, 3, \dots \quad (2.13)$$

This phenomenon is known as photon-number splitting and provides strong evidence for the quantization of the electromagnetic field inside the resonator. It is also an important signature of the strong dispersive regime of cavity quantum electrodynamics in superconducting qubit systems.

The behavior of the system begins to deviate from the ideal dispersive approximation when the number of photons inside the resonator becomes too large. This limit is characterized by the critical photon number,

$$n_{\text{crit}} = \frac{\Delta^2}{4g^2} \quad (2.14)$$

where Δ is the qubit–resonator detuning and g is the coupling strength.

For the parameters used in the present simulation ($g = 100$ MHz, $\Delta = 800$ MHz), the estimated critical photon number is

$$n_{\text{crit}} \approx 16 \quad (2.15)$$

When the intracavity photon population approaches this value, the dispersive approximation begins to break down. Consequently, the spectral peaks broaden and overlap due to increasing nonlinear effects and mixing of dressed-state transitions. In practice, signatures of nonlinear broadening can appear even at lower powers before the theoretical critical photon number is fully reached.

2.6.1 Power Broadening

After studying photon-number splitting, the dependence of the qubit spectral linewidth on the applied drive power was investigated. While photon-number splitting arises due to an increase in the resonator readout power and the resulting intracavity photon population, the qubit spectroscopy linewidth is primarily affected by the strength of the qubit drive tone.

In qubit spectroscopy, the strength of the microwave drive affects the observed linewidth of the transition. At low drive power, the qubit transition appears as a sharp Lorentzian peak, and the linewidth is mainly determined by the intrinsic coherence properties of the qubit.

As the drive power increases, the qubit experiences stronger excitation, which leads to faster transitions between its energy levels. This results in an increase in the effective linewidth of the spectral peak, a phenomenon known as power broadening.

At higher drive powers, the resonance peak becomes broader and less distinct, making precise frequency identification more difficult. In the low-power regime, reliable spectroscopy measurements can therefore be performed without significant distortion of the linewidth.

2.7 Anharmonicity of the Transmon Qubit

Transmon Qubit: Physical Background

A superconducting transmon qubit is a weakly nonlinear quantum oscillator realized using a Josephson junction shunted by a large capacitor. Unlike an ideal harmonic oscillator, the transmon exhibits unequally spaced energy levels due to the nonlinear cosine-shaped potential introduced by the Josephson junction.

Because of this nonlinearity, the system shows weak anharmonicity, which plays a crucial role in enabling selective qubit control by suppressing unwanted transitions to higher energy states.

The lowest energy transitions are:

$$\begin{aligned} f_{01} &: |0\rangle \rightarrow |1\rangle \\ f_{12} &: |1\rangle \rightarrow |2\rangle \end{aligned}$$

In this work, a three-level (qutrit) approximation is used to model the transmon dynamics.

Transmon Hamiltonian

The transmon is described by the Hamiltonian:

$$H = 4E_C(\hat{n} - n_g)^2 - E_J \cos(\hat{\phi})$$

where:

- E_C is the charging energy
- E_J is the Josephson energy
- $\hat{n}$ is the Cooper-pair number operator
- $\hat{\phi}$ is the superconducting phase difference across the Josephson junction

The nonlinear cosine potential leads to non-uniform energy level spacing, resulting in a slight difference between successive transition frequencies.

Anharmonicity of the Transmon

The anharmonicity is defined as:

$$\alpha = f_{12} - f_{01}$$

For a typical transmon qubit, the anharmonicity is negative and small in magnitude. This weak nonlinearity ensures that the system remains close to a harmonic oscillator while still allowing frequency selectivity between transitions, which is essential for qubit operation and readout.

Transmon–Resonator System

The coupled transmon–resonator system is described by the Hamiltonian:

$$H = \hbar\omega_r a^\dagger a + \hbar\omega_q b^\dagger b + \frac{1}{2}\hbar\alpha b^\dagger b^\dagger b b + \hbar\chi a^\dagger a b^\dagger b$$

where:

- ω_r : resonator frequency
- ω_q : qubit frequency
- $a, a^\dagger$: resonator operators
- $b, b^\dagger$: qubit operators
- α : anharmonicity
- χ : dispersive coupling strength

The dispersive coupling strength is given by:

$$\chi = \frac{g^2\alpha}{\Delta(\Delta + \alpha)}$$

where $\Delta = \omega_q - \omega_r$. In the limit $\Delta \gg \alpha$, this reduces to:

$$\chi \approx \frac{g^2}{\Delta}$$

This regime allows the qubit state to shift the resonator frequency without direct energy exchange, forming the basis of dispersive readout in circuit QED systems.

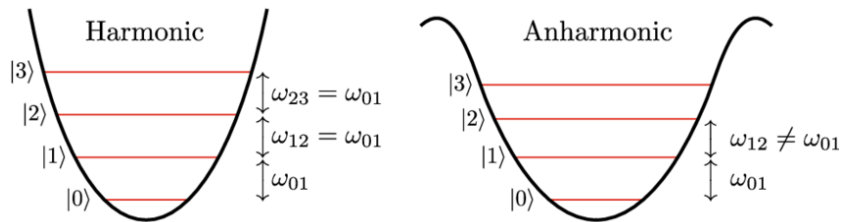


Figure 2.2: Comparison between a harmonic oscillator and a transmon qubit energy spectrum showing unequal level spacing and anharmonicity.

2.8 Time-Domain Characterization of Superconducting Qubits

While frequency-domain spectroscopy provides information about the spectral properties of a superconducting qubit system, time-domain measurements are used to study qubit dynamics and coherence behavior. These experiments investigate the response of the qubit to microwave control pulses and are essential for pulse calibration, coherent control, and characterization of coherence properties in superconducting quantum systems.

Common time-domain experiments include Rabi oscillation measurements, Ramsey interference experiments, and relaxation measurements used to determine coherence times such as T_1 and T_2^* .

2.8.1 Bloch Sphere Representation

The state of a qubit can be visualized using the Bloch sphere representation, which provides a geometric description of a two-level quantum system. A general qubit state can be expressed as

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle \quad (2.16)$$

where θ and ϕ define a point on the surface of the Bloch sphere. The north pole of the Bloch sphere represents the ground state $|0\rangle$, while the south pole represents the excited state $|1\rangle$. Points on the surface correspond to superposition states. Quantum gate operations can be interpreted as rotations on the Bloch sphere. For example, a π -pulse rotates the qubit from the ground state to the excited state, while a $\pi/2$ -pulse creates a coherent superposition state. The Bloch sphere provides a convenient way to visualize qubit rotations and coherent control in time-domain experiments.

2.8.2 Rabi Oscillations

One of the fundamental time-domain experiments used in superconducting qubit systems is the Rabi oscillation experiment. In this experiment, a resonant microwave drive is applied to the qubit while varying either the pulse duration or pulse amplitude. Under resonant driving, the qubit undergoes coherent oscillations between the ground state and the excited state. These coherent oscillations are known as Rabi oscillations. Rabi measurements are important because they allow calibration of control pulses used for qubit manipulation. In particular, the experiment enables determination of:

- the π -pulse, which completely excites the qubit,
- the $\pi/2$ -pulse, which creates a superposition state.

These calibrated pulses form the basis of quantum gate operations in superconducting quantum circuits.

2.8.3 Ramsey Interference and Dephasing Time

The Ramsey interference experiment is used to measure the dephasing properties of a qubit and determine the coherence time T_2^* .

In a Ramsey experiment, two $\pi/2$ -pulses are applied to the qubit with a variable delay between them. During the delay period, the qubit accumulates a relative phase due to the frequency difference between the qubit and the applied drive.

As the delay time increases, interference oscillations known as Ramsey fringes are observed. Due to environmental fluctuations and dephasing mechanisms, the oscillations gradually decay with time. The decay of the Ramsey fringes provides information about the phase coherence properties of the qubit and is used to estimate the effective dephasing time T_2^* . Ramsey measurements provide information about dephasing and frequency fluctuations in superconducting qubits.

2.8.4 Relaxation Time and Energy Decay

Another important time-domain measurement is the relaxation experiment used to determine the energy relaxation time T_1 .

In this experiment, the qubit is first excited using a π -pulse. After excitation, the system is allowed to evolve freely for a variable waiting time before measurement is performed. Over time, the qubit relaxes from the excited state back to the ground state due to energy loss to the surrounding environment. The relaxation process is characterized by the relaxation time T_1 , which describes how long the qubit can remain in the excited state before decaying.

2.8.5 Relaxation and Decoherence

In realistic superconducting systems, qubits continuously interact with their surrounding environment. These interactions lead to decoherence, which limits the performance and reliability of quantum devices.

The relaxation time T_1 describes the loss of energy from the excited state to the ground state, while the dephasing time T_2 characterizes the loss of phase coherence between quantum states.

The coherence times are related through

$$\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\phi} \quad (2.17)$$

where T_ϕ represents the pure dephasing time.

Interactions with the surrounding environment lead to energy relaxation and dephasing, thereby limiting the coherence properties of superconducting qubits.

Chapter 3

Experimental Setup

3.1 Overview of Experimental Setup

The experimental setup used in this work consists of microwave measurement instruments, attenuation stages, and a cryogenic measurement system used for superconducting qubit experiments. The setup is designed to perform resonator and qubit spectroscopy measurements at millikelvin temperatures. Microwave signals are generated at room temperature and sent to the superconducting device through microwave input lines inside the cryostat. The output signals from the device are amplified and measured using microwave measurement instruments.

3.2 Microwave Input Lines and Attenuators

Microwave signals used for resonator and qubit measurements are sent to the device through coaxial cables connected to the cryogenic setup. Since thermal noise from higher-temperature stages can affect the qubit, attenuation stages are added at different stages inside the cryostat. These attenuators help reduce unwanted noise reaching the superconducting device. The attenuation stages additionally help control the microwave power used during spectroscopy measurements.

3.3 Vector Network Analyzer (VNA)

In the measurement setup, the microwave signal generated by the VNA is sent to the resonator-qubit device through the input line inside the cryostat, while the transmitted output signal is collected through the output line and measured at the second port of the VNA.

3.4 Signal Generator (SG)

A microwave signal generator is used to apply the qubit drive tone during spectroscopy measurements. In two-tone spectroscopy experiments, the signal generator sweeps the drive frequency around the qubit transition frequency while the resonator transmission response is continuously monitored.

3.5 Cryogenic Measurement System

Superconducting qubits operate at extremely low temperatures, typically in the millikelvin range, where superconductivity is maintained and thermal excitations become very small. To achieve these temperatures, the qubit device is placed inside a dilution refrigerator, also called a cryostat. The cryostat contains multiple temperature stages that gradually cool the system from room temperature down to nearly 10–20 mK. Operating at millikelvin temperatures is essential because thermal fluctuations at higher temperatures can disturb the qubit states and reduce quantum coherence.

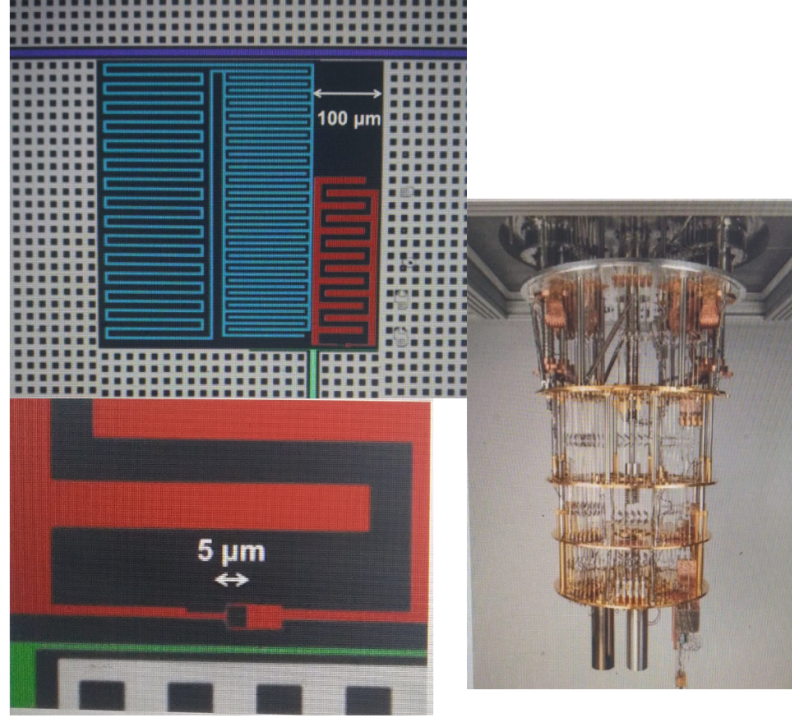


Figure 3.1: Overall experimental setup and cryogenic measurement system used for superconducting qubit characterization.

Chapter 4

Measurement Methodology

4.1 Experimental Methodology

4.1.1 Resonator Spectroscopy Measurements

Resonator spectroscopy measurements were performed using a Vector Network Analyzer (VNA) to characterize the transmission response of the superconducting resonator. The complex transmission parameter S_{21} was measured over a wide frequency range to identify the resonance frequency.

An initial coarse frequency sweep from 5.9 GHz to 6.4 GHz was used to locate the approximate resonance position. The microwave input power was controlled using fixed and digital attenuation in the input line to ensure stable operation of the resonator and to avoid nonlinear effects. The VNA output power was fixed at 0 dBm. The microwave input line included a total attenuation of 76 dB due to cryogenic attenuators and microwave components. Additional digital attenuation was used to control the power reaching the cavity and to avoid nonlinear effects.

The effective cavity power was estimated using the relation

$$P_{\text{cavity}}(\text{dBm}) = 0 - (76 + A_{\text{digital}}) \quad (4.1)$$

The corresponding cavity power values for different digital attenuation settings are listed in Table 4.1.

Table 4.1: Cavity power corresponding to different digital attenuation values.

Digital Attenuation	Total Attenuation	Cavity Power
0 dB	76 dB	-76 dBm
10 dB	86 dB	-86 dBm
50 dB	126 dB	-126 dBm

The measured S_{21} response was further analyzed using Python-based spectroscopy tools and Lorentzian fitting to extract the resonator parameters.

4.1.2 Coarse Frequency Sweep

A coarse frequency sweep was first performed over a wide frequency range to approximately identify the resonance frequency. Due to sparse sampling, this scan provides only a rough estimate with higher noise.

4.1.3 Naive Sweep

In the naive sweep, frequency points are sampled uniformly without adaptive refinement. This method is simple but inefficient for resolving sharp resonance features since most points lie away from resonance.

4.1.4 Adaptive Spectroscopy Algorithm

To improve resolution near resonance, an adaptive sampling method was used after the initial coarse estimate. Frequency points are redistributed around the resonance using a tangent-based scheme:

$$f_{\text{new}} = f_0 + \frac{\kappa}{2} \tan(\theta) \quad (4.2)$$

where f_0 is the estimated resonance frequency and κ is the linewidth. Lorentzian fitting is performed after each iteration to update the resonance parameters.

4.1.5 Signal Processing and Filtering

The measured S_{21} data contains slow background variations and noise. To improve fitting stability, Savitzky–Golay filtering is applied. A coarse window is used for background removal, followed by finer smoothing to suppress noise while preserving the resonance lineshape.

4.1.6 Phase-Asymmetric Lorentzian Fitting

The resonator response is modeled using a phase-asymmetric Lorentzian function to account for experimentally observed asymmetry:

$$S_{21}(f) = A \left[1 - \frac{\text{depth} e^{i\phi}}{1 + 2i(f - f_0)/\kappa} \right] \quad (4.3)$$

where f_0 is the resonance frequency, κ is the linewidth, A is the background amplitude, depth represents the resonance strength, and ϕ is the asymmetry phase. This model captures effects such as impedance mismatch and cable reflections. fitting is performed using nonlinear least squares optimization (`curve_fit` from SciPy) to extract resonance frequency and linewidth.

4.1.7 Readout Power Calibration via Transition Detection

Before performing two-tone qubit spectroscopy, an appropriate readout power must be chosen to avoid nonlinear response of the resonator while maintaining sufficient signal-to-noise ratio.

The optimal readout power was determined by studying the resonator response as a function of input power using VNA-based spectroscopy. For each power level, the transmission parameter S_{21} was measured and fitted with a Lorentzian model to extract the resonance frequency and linewidth.

The power dependence of the resonator exhibited three regimes: low-power linear response, an intermediate transition region, and high-power saturation. To identify the optimal operating point, the response curve was fitted using a sigmoid function, and a knee-point detection method was applied.

The knee-point detection algorithm identifies the point of maximum curvature in the fitted response curve, corresponding to the onset of nonlinear behavior in the resonator.

The optimal readout power was chosen slightly below this identified knee point to ensure stable operation while maintaining sufficient signal-to-noise ratio for spectroscopy measurements.

4.2 Methodology and Simulation

Due to an unexpected warm-up of the dilution refrigerator during the measurement campaign, further qubit-based experimental measurements could not be performed. As a result, the subsequent two-tone spectroscopy and qubit characterization studies were carried out using numerical simulations.

The simulation framework uses simplified parameter values for the superconducting qubit–resonator system for numerical convenience. In particular, parameters such as the resonator frequency are taken as rounded values (6 GHz instead of 6.133 GHz) to simplify the simulation and analysis.

4.2.1 Overview of Simulation Framework

The frequency-domain characterization of the superconducting qubit system was carried out using a modular simulation framework developed in Python. The simulations were performed using QuTiP to model the dynamics of a transmon qubit coupled to a microwave resonator in the circuit QED architecture.

The steady-state response of the coupled system was obtained by solving the Lindblad master equation numerically. From the resulting density matrix, ex-

perimentally measurable quantities such as the transmission coefficient S_{21} were extracted.

The simulation workflow included:

- bare resonator spectroscopy
- coupled qubit–resonator simulation in the dispersive regime
- two-tone spectroscopy
- photon-number splitting analysis
- power broadening analysis
- optimal drive power estimation
- anharmonicity extraction

4.2.2 System Model

The simulated system consists of a superconducting transmon qubit coupled to a microwave resonator. The interaction between the qubit and resonator was modeled using the Jaynes–Cummings Hamiltonian in the dispersive regime.

The coupled system was simulated numerically under different spectroscopy conditions to study the frequency response and power-dependent behavior of the qubit–resonator system.

4.2.3 Simulation Parameters

The simulation parameters used throughout this work are summarized in Table 4.2.

Parameter	Value
Resonator frequency (ω_r)	6.0 GHz
Qubit frequency (ω_q)	5.2 GHz
Coupling strength (g)	100 MHz
Resonator quality factor (Q)	2000
Energy relaxation time (T_1)	1000 ns
Dephasing time (T_2)	1000 ns

Table 4.2: Simulation parameters used for the circuit QED system

4.2.4 Bare Resonator Spectroscopy

The bare superconducting resonator was modeled as a quantum harmonic oscillator, and its frequency response was simulated by applying a microwave drive in the rotating frame. The steady-state solution of the Lindblad master equation was used to compute the system response as a function of drive frequency.

The expectation value of the cavity field was used to extract the transmission coefficient S_{21} , reproducing a standard VNA-like frequency sweep for determining the resonance frequency and linewidth.

4.2.5 Qubit–Resonator System Simulation

The system was then extended to include a superconducting transmon qubit coupled to the resonator, described using the Jaynes–Cummings Hamiltonian in the dispersive regime.

The combined system was modeled using a tensor-product Hilbert space, with dissipation included via collapse operators for resonator decay, qubit relaxation, and dephasing. The steady-state solution of the Lindblad master equation was used to compute the transmission response S_{21} .

4.2.6 Two-Tone Spectroscopy Simulation

To determine the qubit transition frequency, a two-tone spectroscopy simulation was performed. A fixed readout tone was applied at 6.0125 GHz with a readout power of -100 dBm, while a second drive tone was swept across a range of frequencies near the expected qubit transition frequency.

For each drive frequency, the coupled qubit–resonator system was evolved under the system Hamiltonian and the steady-state response was computed numerically using QuTiP. The magnitude of the transmission coefficient, $|S_{21}|$, was recorded as a function of drive frequency.

A distinct peak was observed in the transmission spectrum at the qubit transition frequency, allowing extraction of the qubit resonance from the simulated spectroscopy data.

4.2.7 Photon-Number Splitting Simulation

After determining the qubit transition frequency using two-tone spectroscopy, photon-number splitting was investigated by varying the readout power from -110 dBm to -80 dBm.

For each power level, the transmission spectrum of the coupled qubit–resonator system was computed numerically using QuTiP. As the readout power increased,

additional peaks became visible in the spectroscopy response.

The resonance peaks were identified numerically using the `find_peaks()` function from the SciPy signal-processing library. The evolution and separation of these peaks with increasing readout power were analyzed to study photon-number splitting behavior.

This simulation reproduces the phenomenon of photon-number splitting observed in the strong dispersive regime.

4.2.8 Power Broadening Simulation

Initially, a wide frequency sweep in the range of 5.0–5.5 GHz was performed to capture the overall spectral response of the coupled qubit–resonator system. In this regime, features such as mode splitting and dressed-state signatures were observed due to strong qubit–resonator interaction.

For accurate linewidth extraction, a narrow frequency scan around the qubit transition frequency (5.19–5.21 GHz) was then used. This restricted scan isolates the qubit resonance and produces a well-defined spectral peak suitable for Lorentzian fitting.

To study power broadening, power-dependent spectroscopy simulations were performed. In this study, the qubit drive frequency was swept while the drive power was systematically varied.

For each power level, the system dynamics were simulated by solving the Lindblad master equation using QuTiP, and the corresponding transmission spectrum was computed.

At low drive powers, the spectrum exhibits a sharp Lorentzian peak. As the drive power increases, the peak broadens due to increased transition rates induced by stronger driving. This evolution of linewidth with drive power is used to characterize power broadening in the system.

4.2.9 Optimal Power Estimation

In qubit spectroscopy, very low drive powers lead to poor signal-to-noise ratio, while high drive powers introduce significant power broadening effects. Therefore, an optimal drive power regime is required in which the qubit resonance remains clearly resolvable without excessive linewidth distortion.

To identify this operating point, a data-driven optimization approach was implemented. The qubit drive power was varied systematically, and for each value, the spectral response was fitted using a Lorentzian function to extract both the peak height and the full width at half maximum (FWHM).

An optimal power threshold was defined by imposing a constraint on the linewidth, such that it remains within a controlled range relative to the low-power (natural) linewidth. A binary search algorithm was then used to efficiently determine the maximum drive power that satisfies this condition.

The binary search procedure can be summarized as follows:

- Initialize the power range $[P_{\min}, P_{\max}]$
- Compute the midpoint drive power
- Evaluate the corresponding linewidth using Lorentzian fitting
- If the linewidth is below the threshold, the search is shifted toward higher power
- Otherwise, the search is shifted toward lower power
- Repeat this process until convergence is achieved

This iterative approach ensures efficient convergence to the optimal operating point without requiring an exhaustive sweep over the entire power range.

The resulting optimal drive power provides a balance between sufficient signal visibility (peak height) and minimal power broadening (linewidth stability). This optimized point is subsequently used for further time-domain analysis, including the extraction of coherence properties such as T_2^* .

4.2.10 Transmon Anharmonicity Simulation

To account for higher excited-state transitions, the transmon qubit was modeled as a weakly anharmonic multi-level quantum system rather than an ideal two-level system.

Driven System and Rotating Frame Representation

To simulate spectroscopy under external microwave drives, the system is analyzed in a rotating frame defined by the applied drive frequencies. This removes fast oscillating terms and simplifies numerical computation.

The effective Hamiltonian is given by:

$$H_{\text{eff}} = \hbar(\omega_r - \omega_c)a^\dagger a + \hbar(\omega_q - \omega_d)b^\dagger b + \frac{1}{2}\hbar\alpha b^\dagger b^\dagger b b + \hbar\chi a^\dagger a b^\dagger b + H_{\text{drive}}$$

where the drive term is:

$$H_{\text{drive}} = \epsilon_r(a + a^\dagger) + \epsilon_q(b + b^\dagger)$$

Here:

- ϵ_r : corresponds to readout tone
- ϵ_q : corresponds to qubit spectroscopy drive

This formulation is used in numerical simulations to compute the steady-state response under two-tone spectroscopy conditions.

Spectroscopy Simulation Method

Two-tone spectroscopy is performed by sweeping the drive frequency f_d over the range 4.8–5.4 GHz. For each frequency:

- Hamiltonian is constructed
- steady-state density matrix is computed
- cavity expectation value $\langle a \rangle$ is extracted

The transmission is calculated as:

$$S_{21} = 1 - \frac{\kappa}{i 2\epsilon_r} \langle a \rangle$$

Brute-Force Spectroscopy Method

In the brute-force approach, the entire frequency range is sampled using a uniform grid (frequency scan) with fixed resolution. No prior information about resonance locations is used.

This method provides complete spectral coverage but is computationally expensive due to redundant sampling in off-resonant regions.

Adaptive Spectroscopy Algorithm

To improve computational efficiency, a multi-stage adaptive spectroscopy approach was implemented. A coarse scan is first used to identify approximate resonance regions, followed by local refinement around the extracted f_{01} transition. Using the anharmonicity parameter, the second transition is estimated as $f_{12} \approx f_{01} + \alpha$, and a targeted scan is performed around this predicted region.

This adaptive strategy significantly reduces the number of required frequency points while maintaining accurate identification of qubit transitions.

4.3 Time-Domain Measurement Simulations

4.3.1 Rabi Oscillation Simulation

Rabi oscillations were simulated to calibrate qubit control pulses in the circuit QED system. The driven qubit–resonator dynamics were modeled using the Lindblad master equation implemented in QuTiP, including qubit, resonator, and microwave drive terms. The measurement response was obtained using input–output theory in the dispersive regime, where the transmission parameter S_{21} was computed as a function of drive pulse duration. This produces coherent oscillations between the ground and excited states.

The oscillations were fitted using a damped cosine function:

$$f(t) = Ae^{-t/\tau} \cos(2\pi ft + \phi) + C \quad (4.4)$$

The extracted oscillation frequency was used to determine the π -pulse duration for further simulations.

4.3.2 Ramsey Interference Simulation

Ramsey interference was simulated to extract the coherence time T_2^* . The protocol uses two $\pi/2$ pulses separated by a variable delay, during which the qubit accumulates a relative phase due to detuning. The resulting Ramsey fringes decay due to dephasing and are modeled as:

$$f(t) = Ae^{-t/T_2^*} \cos(2\pi f_{\text{det}}t + \phi) + C \quad (4.5)$$

The extracted T_2^* quantifies phase decoherence due to environmental noise and fluctuations.

4.3.3 T_1 Relaxation Measurement

Energy relaxation time T_1 was simulated using a pulse-and-wait protocol. The qubit is first excited using a calibrated π -pulse, followed by free evolution for varying delay times. The excited-state population decays exponentially as:

$$f(t) = Ae^{-t/T_1} + C \quad (4.6)$$

The fitted T_1 characterizes the energy loss timescale of the qubit due to coupling with the environment.

Chapter 5

Results and Discussion

5.1 Resonator Spectroscopy Results

5.1.1 Coarse and Adaptive Resonator Spectroscopy

Resonator spectroscopy measurements were performed at different readout power levels using a Vector Network Analyzer (VNA). Coarse frequency sweeps were first used to approximately locate the resonance, followed by adaptive sweeps with higher resolution near the resonance dip.

The extracted resonance parameters obtained from phase-asymmetric Lorentzian fits are summarized in Table 5.1. The adaptive scans provide improved resolution around the resonance compared to the coarse measurements.

Table 5.1: Extracted resonator parameters from phase-asymmetric Lorentzian fits at different readout power levels.

Power (dB)	Resonance Frequency (GHz)	κ (MHz)
0	6.163615	0.359953
5	6.163619	0.366587
10	6.163655	0.332550
15	6.163389	0.380703
20	6.162965	0.461551
25	6.162725	0.457162
30	6.162706	0.472047
35	6.162674	0.494412
40	6.162746	0.484457
45	6.162623	0.699648
50	6.162661	0.619832

Overall, the results show good agreement with the phase-asymmetric Lorentzian model across all power levels.

5.1.2 Power Dependence of Resonator Response

A systematic change in the resonator response was observed as the readout power was varied.

At lower powers, the resonance dip appeared broader and relatively shallow. As the input power increased, the resonance became sharper and more well-defined. This behavior is consistent with the reduction of dielectric loss mechanisms at higher drive powers.

The extracted linewidth values showed noticeable power dependence, indicating changes in the effective dissipation of the resonator with increasing microwave power.

Overall, the adaptive spectroscopy and fitting procedure enabled stable extraction of resonator parameters across different operating powers.

5.1.3 Observation of Photon-Number Splitting

Photon-number splitting was observed in the simulated two-tone spectroscopy spectra as a function of increasing readout power. At low readout power (-110 dBm), two well-resolved peaks were observed with a frequency separation close to $2\chi \approx 25$ MHz.

As the readout power was gradually increased, the average photon number in the resonator increased, leading to the appearance of additional spectral peaks corresponding to higher photon-number states. This evolution indicates the progressive occupation of higher Fock states in the resonator field.

Readout Power	Avg Photon Number	No. of Peaks	Peak Spacing
-110 dBm	0.01	2	24.8 MHz
-105 dBm	0.03	2	24.8 MHz
-100 dBm	0.099	2	24.8 MHz
-95 dBm	0.32	3	24.8 MHz
-90 dBm	0.99	4	24.8 MHz
-85 dBm	3.08	5	~ 24 MHz
-80 dBm	4.77	merged	nonlinear

Table 5.2: Power dependence of photon-number splitting in two-tone spectroscopy

The increase in the number of resolved peaks with power clearly demonstrates photon-number splitting in the strong dispersive regime.

Equal Peak Spacing

The observed peak spacing of ~ 24.8 MHz is consistent with the theoretical dispersive prediction 2χ , where $\chi = 12.5$ MHz, confirming the expected photon-number-dependent frequency shifts in the dispersive regime.

High-Power Nonlinear Regime

At high readout power (-80 dBm), the spectroscopy spectrum no longer exhibited well-resolved peaks. Instead, broad and overlapping spectral features were observed, indicating the onset of nonlinear effects and partial breakdown of the simple dispersive approximation.

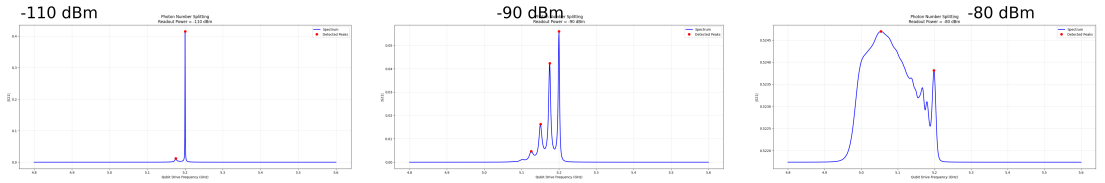


Figure 5.1: Evolution of photon-number splitting in two-tone spectroscopy with increasing readout power. Low-power spectra show a small number of resolved photon-number peaks, while intermediate powers reveal multiple equally spaced transitions separated by approximately 2χ . At high power, nonlinear broadening and overlap of spectral features indicate the onset of breakdown of the simple dispersive approximation.

5.1.4 Power Broadening Analysis

Linewidth Evolution with Drive Power ($g = 10$ MHz)

The evolution of the qubit spectral linewidth with increasing drive power for the weak coupling regime ($g = 10$ MHz) is summarized in Table 5.3. The resonance frequency remains approximately constant, while the linewidth increases gradually at low power and rapidly at higher drive strengths.

The results clearly show a monotonic increase in linewidth with drive power, consistent with the expected power broadening behavior.

Strong Coupling Regime Comparison ($g = 100$ MHz)

In the strong coupling regime ($g = 100$ MHz), the linewidth increases much more rapidly with drive power, as shown in Table 5.4. At high powers (beyond approximately -85 dBm), the spectrum becomes strongly broadened and Lorentzian fitting becomes unreliable due to nonlinear saturation effects.

This indicates the onset of strong nonlinear effects and saturation-induced breakdown of simple Lorentzian behavior at high drive powers.

Qubit Drive Power (dBm)	Resonance Frequency (GHz)	FWHM Linewidth (MHz)
-130	5.199974	0.333
-125	5.199974	0.346
-120	5.199974	0.386
-115	5.199973	0.491
-110	5.199970	0.730
-105	5.199962	1.205
-100	5.199946	2.081
-95	5.199928	3.653
-90	5.199914	6.457
-85	5.199907	11.455
-80	5.199903	20.351
-75	5.199901	36.185
-70	5.199901	64.402
-65	5.199900	114.902

Table 5.3: Linewidth evolution in the weak coupling regime ($g = 10$ MHz)

Qubit Drive Power (dBm)	Resonance Frequency (GHz)	FWHM Linewidth (MHz)
-130	5.199986	0.623
-125	5.199986	0.637
-120	5.199986	0.679
-115	5.199984	0.797
-110	5.199978	1.091
-105	5.199960	1.723
-100	5.199900	2.933
-95	5.199711	5.111
-90	5.199108	10.215
-85	5.197885	34.660
-80	5.196423	2719.058
-75	5.194620	3512.857
-70	Fit Failed	—
-65	Fit Failed	—

Table 5.4: Linewidth evolution in the strong coupling regime ($g = 100$ MHz)

Comparison Between Coupling Regimes

Strong Coupling ($g = 100$ MHz)	Weak Coupling ($g = 10$ MHz)
Strong nonlinear effects at lower power	Smooth linewidth evolution
Rapid saturation	Gradual broadening
Unstable fitting at high power	Reliable Lorentzian fits
Breakdown of simple model	Well-behaved response

Table 5.5: Comparison between weak and strong coupling regimes

The weak coupling regime provides a stable and well-defined Lorentzian response over a wide range of drive powers, making it suitable for accurate linewidth extraction. In contrast, the strong coupling regime exhibits faster onset of nonlinearities, limiting the valid fitting range.

Scan Range Dependence

For linewidth extraction and power broadening analysis, a narrow frequency scan around the qubit transition (5.19–5.21 GHz) was used. This regime isolates the f_{01} transition and produces well-defined Lorentzian peaks suitable for fitting.

In contrast, a wide frequency scan (5.0–5.5 GHz) reveals the full coupled qubit–resonator spectrum, including dressed-state features and spectral splitting. However, due to the presence of hybridized modes, Lorentzian fitting becomes unreliable in this regime.

Therefore, the narrow scan is the appropriate choice for power broadening analysis, as it directly captures the evolution of the qubit linewidth.

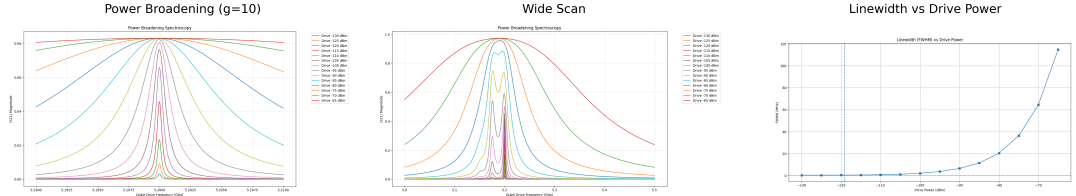


Figure 5.2: Power broadening effects and linewidth dependence on drive power showing spectral broadening and transition saturation behavior.

5.1.5 Power Dependence of Peak Height and Linewidth

The dependence of peak height and linewidth on qubit drive power is summarized in Table 5.6. As the drive power increases, the peak height initially increases and then saturates, while the linewidth increases monotonically due to power broadening effects.

The natural linewidth at low drive power is found to be:

$$\Gamma_0 = 0.3329 \text{ MHz}$$

A threshold criterion of:

$$\Gamma_{\text{allowed}} = 1.2 \times \Gamma_0$$

was used to define the acceptable broadening limit.

Power (dBm)	Peak Height	Linewidth (MHz)
-130.00	0.003172	0.3329
-125.00	0.009259	0.3465
-120.00	0.023552	0.3863
-115.00	0.046022	0.4915
-110.00	0.065934	0.7301
-105.00	0.076480	1.2049
-100.00	0.080750	2.0814
-95.00	0.082359	3.6527
-90.00	0.082889	6.4566
-85.00	0.083035	11.4546
-80.00	0.083081	20.3511
-75.00	0.083131	36.1849
-70.00	0.083298	64.4020
-65.00	0.083856	114.9022

Table 5.6: Power dependence of peak height and linewidth in qubit spectroscopy.

The binary search algorithm yields an optimal drive power of:

$$P_{\text{opt}} \approx -119.06 \text{ dBm}$$

At this operating point, the system provides a good compromise between signal visibility and minimal linewidth distortion.

Physical Interpretation

At low drive powers, the signal is weak but the spectral peak remains sharp. As the power increases, the signal improves, but nonlinear effects begin to broaden the resonance. Beyond a certain power, power broadening dominates and reduces spectral resolution.

The optimal drive power therefore represents a balance point between measurement sensitivity and spectral accuracy.

5.1.6 Anharmonicity Results

Low Drive Regime

At low drive power (-100 to -90 dBm), the system remains in the linear regime. Only the fundamental transition f_{01} is observed near 5.2 GHz. The f_{12} transition is not visible due to negligible population in the first excited state.

Intermediate Drive Regime

At moderate drive power (-85 to -80 dBm), population begins to transfer into the $|1\rangle$ state. As a result, the f_{12} transition starts to emerge. Slight broadening of the f_{01} peak is also observed due to increased drive strength.

High Drive Regime

At high drive power (-70 dBm), strong multi-level effects appear. The f_{01} transition exhibits significant power broadening, while the f_{12} transition becomes clearly visible. The extracted anharmonicity is approximately:

$$\alpha \approx -300 \text{ MHz}$$

Comparison: Brute Force vs Adaptive Spectroscopy

Both brute-force and adaptive methods produce identical spectral features, confirming the correctness of the adaptive algorithm.

- Brute-force method scans the entire frequency range uniformly
- Adaptive method focuses only on physically relevant regions

This makes adaptive spectroscopy more efficient for real experimental quantum systems.

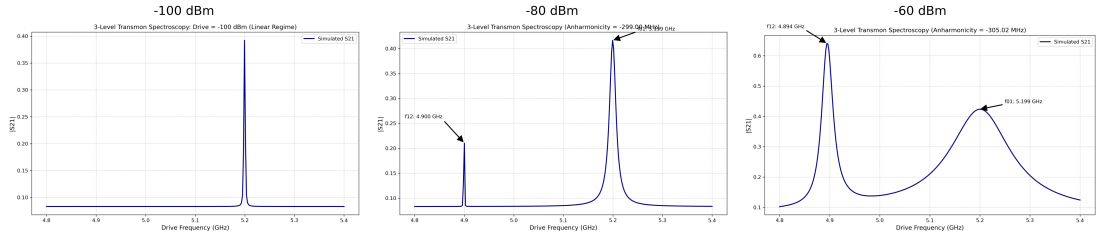


Figure 5.3: Brute-force spectroscopy results at low (-100 dBm), intermediate (-80 dBm), and high (-60 dBm) drive powers. The higher-level f_{12} transition gradually emerges with increasing drive power, while strong power broadening appears at high drive strength.

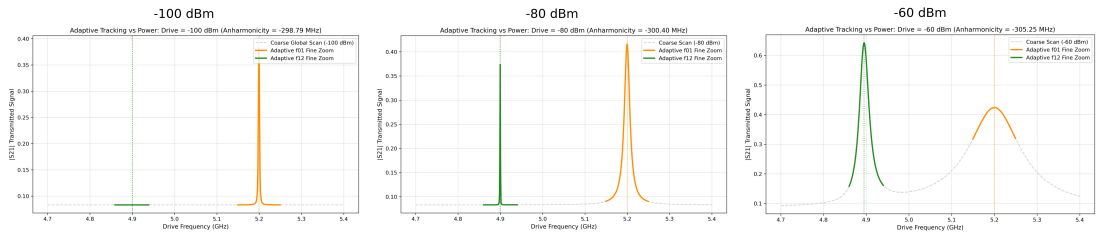


Figure 5.4: Adaptive spectroscopy results at low (-100 dBm), intermediate (-80 dBm), and high (-60 dBm) drive powers, showing efficient resolution of the f_{01} and f_{12} transitions with increasing drive strength.

Results Summary

Uniform (Brute Force) Spectroscopy Results

Drive Power (dBm)	Method	f_{01} (GHz)	f_{12} (GHz)	α (MHz)	Physical Regime
-100	Uniform Sweep	5.1993	Not Detected	N/A	Ultra-low
-95	Uniform Sweep	5.1993	Not Detected	N/A	Linear response in
-90	Uniform Sweep	5.1993	Below Threshold	N/A	Weak f_{12}
-85	Uniform Sweep	5.1993	Below Threshold	N/A	Onset of
-80	Uniform Sweep	5.1993	4.9003	-299.00	Multi-level regime
-75	Uniform Sweep	5.1993	4.9003	-299.00	Stable p
-70	Uniform Sweep	5.1993	4.8983	-301.00	Power bro
-65	Uniform Sweep	5.1993	4.8983	-301.00	Severe sa
-60	Uniform Sweep	5.1993	4.8943	-305.02	Dressing

Table 5.7: Uniform spectroscopy results for different drive powers

Adaptive Spectroscopy Results

Power (dBm)	f_{01}^{uni}	f_{01}^{ad}	f_{12}^{uni}	f_{12}^{ad}	α^{uni}	α^{ad}	Insight
-100	5.1993	5.1990	Not Detected	4.9002	N/A	-298.79	Weak tran
-95	5.1993	5.1990	Not Detected	4.9002	N/A	-298.79	High-resolu
-90	5.1993	5.1990	Below Threshold	4.9002	N/A	-298.79	Low-signal
-85	5.1993	5.1990	Below Threshold	4.9002	N/A	-298.79	Nonlinear o
-80	5.1993	5.2000	4.9003	4.8996	-299.00	-300.40	Multi-level solved
-75	5.1993	5.2000	4.9003	4.8996	-299.00	-300.40	Stable excit
-70	5.1993	5.2000	4.8983	4.8988	-301.00	-301.21	Power bro
-65	5.1993	5.2000	4.8983	4.8980	-301.00	-302.02	Strong satu

Table 5.8: Comparison of brute-force and adaptive spectroscopy results

The spectroscopy results demonstrate that the adaptive method successfully resolves weak higher-level transitions that are difficult to detect using uniform frequency sweeps. The extracted anharmonicity remains consistently close to -300 MHz across different drive powers, confirming the weakly anharmonic nature of the transmon qubit. In addition, the adaptive approach achieves comparable spectral accuracy while significantly reducing unnecessary frequency sampling.

Chapter 6

Conclusion and Future Work

This thesis presented a simulation-based study of a superconducting transmon qubit coupled to a microwave resonator in the circuit QED framework using numerical simulations based on the Lindblad master equation. In the frequency domain, resonator and qubit spectroscopy were analyzed using brute-force and adaptive methods. The resonator response was modeled using a phase-asymmetric Lorentzian function, enabling extraction of key parameters such as resonance frequency and linewidth. Photon-number splitting was observed in qubit spectroscopy, confirming strong dispersive coupling. Power broadening effects were also studied, showing linewidth with increasing drive power and nonlinear system response. In the time domain, Rabi oscillations and Ramsey interference were simulated to extract coherence properties. The relaxation time (T_1) and dephasing time (T_2^*) were obtained from exponential decay fits, consistent with expected decoherence behavior. Optimization methods such as knee-point detection for readout power and binary search for optimal drive power were implemented to ensure stable spectroscopy operation. Overall, the study provides a numerical framework for analyzing transmon qubit dynamics and highlights the trade-off between measurement accuracy, computational efficiency, and nonlinear quantum effects in circuit QED systems.

6.1 Future Work

Future work will focus on experimental validation of the developed spectroscopy protocols on real superconducting hardware and comparison with simulation results. The study can be extended to multi-qubit systems to investigate coupled qubit dynamics. Further work may include incorporation of realistic noise models, decoherence channels, and development of quantum error correction and mitigation techniques in superconducting architectures.

Appendix A

Additional Methods and Algorithms

A.1 Knee Point Detection for Optimal Readout Power

The power-dependent resonator response was fitted using a sigmoid function to model the transition behavior. A knee-point detection method was applied to the fitted curve, where the point of maximum curvature was identified. This corresponds to the onset of nonlinear behavior in the resonator response. The optimal readout power was chosen slightly below this knee point to ensure stable operation while maintaining a good signal-to-noise ratio for spectroscopy measurements.

A.2 Binary Search for Optimal Drive Power

A binary search algorithm was used to determine the maximum drive power satisfying a linewidth constraint.

- Initialize power range $[P_{\min}, P_{\max}]$
- Compute midpoint power P_{mid}
- Evaluate linewidth using Lorentzian fitting
- If linewidth $< \Gamma_{\text{threshold}}$, set $P_{\min} = P_{\text{mid}}$
- Else, set $P_{\max} = P_{\text{mid}}$
- Repeat until $|P_{\max} - P_{\min}|$ is below tolerance

A.3 Brute Force Spectroscopy Method

In the brute-force approach, the entire frequency or power range is sampled uniformly without any prior knowledge of resonance locations. While this method ensures complete spectral coverage, it is computationally expensive due to redundant sampling in off-resonant regions. This method serves as a baseline for comparison with more efficient adaptive techniques.

A.4 Adaptive Spectroscopy Algorithm

The adaptive spectroscopy method was implemented to reduce computational cost while maintaining spectral accuracy. The procedure consists of the following steps:

- A coarse frequency scan is performed to identify approximate resonance locations.
- A refined scan is applied around the f_{01} transition.
- The second transition f_{12} is estimated using the anharmonicity relation.
- A targeted high-resolution scan is performed around the predicted f_{12} region.

This adaptive approach significantly reduces unnecessary sampling in off-resonant frequency regions while preserving accuracy.

A.5 Lorentzian Fitting Model

The resonator and qubit spectra were fitted using a Lorentzian model of the form:

$$S_{21}(f) = A \left(1 - \frac{\text{depth}}{1 + 2i(f - f_0)/\kappa} \right)$$

This model was used to extract resonance frequency, linewidth, and amplitude parameters.

A.6 Simulation Framework Details

The simulations were performed using the QuTiP library by solving the Lindblad master equation in steady state using the `mesolve` solver. Collapse operators included:

- resonator decay (κ)

- qubit relaxation (T_1)
- dephasing (T_2)

A.7 Time Domain Results

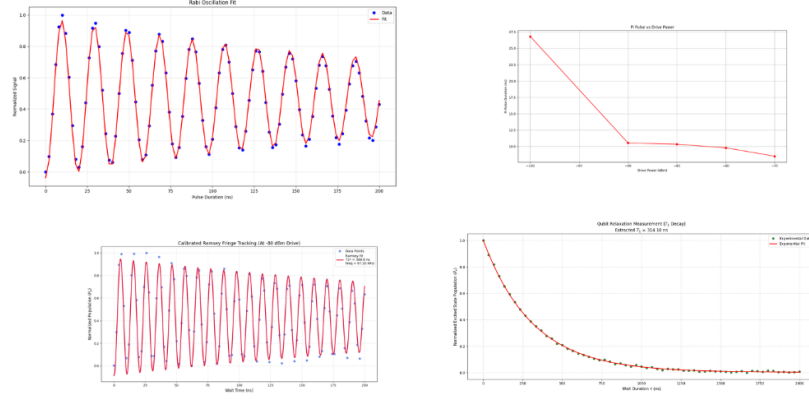


Figure A.1: Additional time-domain measurement results obtained from the transmon qubit simulations.

Bibliography